

GEMINI PRE-PRODUCTION RESEARCH

BRIEF: TASCAM Sonicview Mixer Series

STAGE 1 — Product Identification and Technical Verification

The primary objective of this stage is to establish the precise architectural reality of the TASCAM Sonicview series, strictly relying on official technical documentation and verifiable engineering specifications. The research explicitly isolates the Sonicview line from the previously researched TASCAM Model series and Studio Bridge architectures, confirming that Sonicview represents a fundamentally distinct category of digital audio processing and network routing designed for mission-critical deployment.

Confirmed Lineup and Legacy Exclusions

The verified TASCAM Sonicview series currently consists of two primary console form factors and one dedicated networked stagebox, with console variations defined by power supply topology and included expansion media. The research process explicitly excluded legacy digital consoles, such as the DM-4800, to focus strictly on the current Field Programmable Gate Array (FPGA) ecosystem¹. The confirmed lineup for this production brief is detailed in the table below.

Confirmed Model	Channel/Fader Count	Power Topology	Verification Source
TASCAM Sonicview 16XP	16+1 motorized faders	Single AC power, IF-MTR32 recording card included	VERIFIED ³
TASCAM Sonicview 16dp	16+1 motorized faders	Dual redundant (AC + DC), IF-MTR32 optional depending on region	VERIFIED ⁴
TASCAM Sonicview 24XP	24+1 motorized faders	Single AC power, IF-MTR32 recording card included	VERIFIED ¹
TASCAM Sonicview 24dp	24+1 motorized faders	Dual redundant (AC + DC), IF-MTR32 optional depending	VERIFIED ²

		on region	
TASCAM SB-16D	16-in/16-out Stagebox	Internal AC + Redundant DC Input	VERIFIED ⁸

Architectural Relationship to Prior TASCAM Series

It is verified that the Sonicview series shares absolutely no signal-path architecture with the previously researched TASCAM Model series (Model 12/16/24/2400) or Studio Bridge¹. The Model series utilizes a hybrid architecture featuring an analog summing bus with parallel digital capture. In stark contrast, Sonicview is a native digital mixing console. In this specific technical context, a "digital mixing console" dictates that all analog input signals are immediately digitized at the physical edge of the chassis via 32-bit Analog-to-Digital Converters (ADCs), mathematically summed and processed exclusively within an internal high-speed FPGA, and subsequently routed either via deterministic Audio-over-IP (AoIP) networks or converted back to analog via 24-bit Digital-to-Analog Converters (DACs)¹. VERIFIED.

Inter-Lineup Relationship

The investigation into the structural relationship between the models within the Sonicview lineup reveals a shared core. The Sonicview 16 (XP/dp) and Sonicview 24 (XP/dp) possess an identical internal digital routing matrix, DSP engine, bus count, processing capabilities, and dual-partition operating system. The models are genuinely differentiated solely by their physical Human-Machine Interface (HMI) footprint and the scale of their local analog I/O⁵. The Sonicview 16 utilizes a 16+1 motorized fader bank and dual 7-inch touchscreens, yielding 16 local XLR analog inputs; the Sonicview 24 scales this to a 24+1 motorized fader bank and triple 7-inch touchscreens, yielding 24 local XLR analog inputs¹. Regardless of the physical chassis size, the internal digital routing matrix remains locked at an identical 44 input channels (40 mono, 2 stereo) for all variants¹. VERIFIED.

Core Digital Architecture and DSP Engine

The computational heart of the Sonicview series is a 54-bit floating-point FPGA mixing engine, representing a severe departure from older, fixed-point DSP chip architectures¹. In digital signal processing theory, floating-point mathematics allows the decimal point to "float" dynamically based on the amplitude of the signal. The Sonicview allocates 42 bits for actual audio amplitude data and reserves 12 bits exclusively for dynamic headroom¹³. This specific bit-allocation strategy yields an internal dynamic range that far exceeds the physical limitations of analog circuitry, meaning it is mathematically impossible to internally clip the summing bus, even when aggregating massive channel counts with overlapping transient peaks. Furthermore, the FPGA architecture facilitates massively parallel processing, resulting in an ultra-low internal processing latency of exactly 2 samples, equating to 20.8 microseconds at the system's native 96 kHz continuous sampling rate. The total analog-to-analog roundtrip latency through the entire A/D, DSP, and D/A sequence is 0.51 milliseconds, ensuring pristine phase coherence for

critical live in-ear monitoring applications⁴. VERIFIED.

Full Channel Count and I/O Configuration

The signal routing matrix is consistent across the product line, heavily utilizing flexible busing structures.

- **Internal Channels:** The FPGA accommodates 44 total input channels consisting of 40 discrete mono channels and 2 stereo pairs, supplemented by 4 dedicated stereo FX returns².
- **Mix Buses:** The architecture provides 22 flexible mix buses that engineers can freely configure as subgroups, auxiliary sends, or matrix outputs. This is augmented by a main L/R stereo bus and 4 dedicated FX send buses, culminating in a 32-bus processing infrastructure¹.
- **Sonicview 16 Analog I/O:** The 16-channel physical variant provides 16 XLR mic/line inputs, with channels 9 through 16 configured as combo XLR/TRS inputs to accommodate line-level instruments. It features 16 XLR line outputs, 2 RCA stereo auxiliary inputs, 2 TRS insert points (on channels 7/8), 1 XLR talkback input, and 2 XLR monitor outputs⁴.
- **Sonicview 24 Analog I/O:** The expanded 24-channel variant provides 24 XLR mic/line inputs, with channels 17 through 24 configured as combo XLR/TRS inputs. Output and auxiliary routing mirrors the 16-channel model, providing 16 XLR line outputs, 2 RCA stereo auxiliary inputs, 2 TRS insert points (on channels 15/16), 1 XLR talkback input, and 2 XLR monitor outputs¹. VERIFIED.

Preamplifier Specifications

The Sonicview series introduces the TASCAM Class 1 HDIA (High Definition Instrumentation Architecture) preamplifier¹. It is explicitly verified that this is a completely different design from the Ultra-HDDA preamps utilized in the Model series. The HDIA design is rooted in true instrumentation amplifier topologies at the very first gain stage. Instrumentation amplifiers inherently possess an exceptionally high input impedance and superior Common-Mode Rejection Ratio (CMRR), making them highly immune to induced electromagnetic interference over long cable runs. TASCAM utilized discrete components, including oversized capacitors, to ensure the power supply delivers sufficient instantaneous energy to cleanly pass low-frequency, high-amplitude transients¹³. The measurable engineering result is an Equivalent Input Noise (EIN) rated at -128 dBu or less, and a maximum input handling capability of +32 dBu (with the physical pad engaged and trim minimized)⁴. These preamps feed directly into 32-bit ADCs operating at 96 kHz⁴. VERIFIED.

Control Surface Architecture

The human-machine interface revolves around the TASCAM VIEW (Visual Interactive Ergonomic Workflow) system¹. Depending on the model, the surface features either two or three 7-inch color resistive touchscreens. The VIEW architecture allows operators to select from three distinct cognitive layouts: Channel Strip View (monitoring eight adjacent channels mimicking an analog desk), Module View (monitoring three processing blocks for a single selected channel), and Individual View (a fully customized monitoring array)¹. The tactile layer features 100mm motorized faders (16+1 or 24+1) that map to custom programmable layers,

facilitating advanced functions like "Sends on Fader" and DCA spill¹. Furthermore, 18 assignable physical User Keys with full-color LED feedback allow instant recall of tap tempo, mute groups, and snapshot state changes⁷. VERIFIED.

Onboard DSP and Mixing Architecture

The internal processing capability scales uniformly across all channels without compromising the DSP pool. Every input channel includes an initial Delay stage, Phase inversion, Digital Trim, a High-pass Filter, a dynamics block featuring a Gate/Expander/De-esser, a 4-band Parametric EQ (PEQ), and a secondary dynamics block for a Compressor/Ducker². On the output side, all 22 flex buses are equipped with a 31-band Graphic EQ (GEQ), a Real-Time Analyzer (RTA), an additional 4-band PEQ, a Compressor/Ducker, and Delay circuitry¹. The console features full scene/snapshot recall capabilities—a foundational feature for digital broadcast and touring consoles that allows instantaneous recall of preamp gain, digital routing matrices, fader positions, and all DSP parameters⁷. VERIFIED.

Recording and Multitrack Capability

Unlike the Model series which features a primary integrated SD recorder as part of its hybrid nature, Sonicview is heavily oriented toward USB interfacing and expansion card modularity. The console natively features an integrated 32-in/32-out asynchronous USB audio interface supporting up to 32-bit/96 kHz operation for deep DAW integration¹. Multitrack recording to physical media is handled via the IF-MTR32 expansion card (standard on "XP" variants). This card executes 32-track direct-to-SDXC recording at 48 kHz (or 16-track at 96 kHz) and includes an automated file-save state that closes the write-file every 60 seconds to prevent data loss during sudden power failures⁴. Direct 2-track stereo recording to a top-panel USB flash drive is also supported¹. VERIFIED.

Networking, Remote Control, and Stagebox Integration

The Sonicview ecosystem is fundamentally an Audio-over-IP (AoIP) routing node. The chassis incorporates a native 64-in/64-out Dante interface (operating at 64x64 at 48 kHz, or 32x32 at 96 kHz). The interface utilizes Primary and Secondary etherCON ports to support SMPTE ST 2022-7 standard redundancy⁴. In this mode, two identical packet streams are transmitted across distinct physical networks; if one network fails, the receiver seamlessly reconstructs the audio from the secondary stream with zero sample loss.

For high-level broadcast integration, the console can be expanded with the IF-ST2110 card, providing 64x64 SMPTE ST 2110-30/31 compatibility and NMOS IS-04/05 discovery for integration into IP-based television video/audio fabrics¹. The operating system natively supports the Ember+ remote control protocol, enabling newsroom automation systems (like Viz Mosart or Nxtedition) to directly drive the console's faders and DSP without human intervention²⁰. Remote operation by human engineers is facilitated via the TASCAM Sonicview Control application (Windows, macOS, iPadOS) connected via a dedicated Ethernet control port⁴. VERIFIED.

Physical Form Factor, Dimensions, and Power Topologies

The physical dimensions scale predictably with the fader count.

- **Sonicview 16 (XP/dp):** Dimensions: 472.0(W) x 228.1(H) x 554.4(D) mm. Weight: 13 kg. Power Consumption: 65W⁴.
- **Sonicview 24 (XP/dp):** Dimensions: 690.8(W) x 228.1(H) x 554.4(D) mm. Weight: 18 kg. Power Consumption: 85W².

The critical distinction in power topology lies in the "dp" (Dual Power) models. While the XP models rely on a standard single AC input (100-240V), the dp models feature an integrated AC supply and an additional 4-pin XLR DC input. By connecting the TASCAM PS-P2450 AC/DC external adapter (delivering DC24V, 5.0A) to this secondary port, the console achieves true redundant power operation. Should the primary AC circuit fail, the console automatically and seamlessly draws power from the DC input without dropping a single packet of audio⁶.

VERIFIED.

STAGE 2 — Buyer Psychology and Technical Motivation

The buyer psychology for the TASCAM Sonicview series diverges entirely from the tactile-nostalgia, "escape-the-DAW" mindset that characterizes the Model series user. The Sonicview buyer is a systems architect, a broadcast engineer, or a technical director operating in mission-critical environments where determinism, scalability, and absolute reliability are the primary currencies. Their technical motivations are rooted in solving complex signal routing challenges in high-stakes environments—specifically outside broadcast (OB) vehicles, large-scale houses of worship, corporate AV hubs, and live touring rigs¹⁴.

Deterministic IP Audio and Infrastructure Interoperability

The modern audio professional at this tier is actively migrating from point-to-point analog multicore or AES3 cabling toward Audio-over-IP (AoIP) fabrics. The Sonicview buyer is not looking for a standalone desk; they are looking for a network node. The native inclusion of Dante 64x64 with ST 2022-7 redundancy satisfies the core requirement for scalable, loss-proof live audio¹⁴. Furthermore, the optional inclusion of SMPTE ST 2110 via the IF-ST2110 card serves a highly specific, elite buyer in the broadcast sector who must interface audio directly into IP video network fabrics. The motivation here relies on NMOS IS-04 (Discovery and Registration) and IS-05 (Connection Management) to ensure the audio console can be automatically identified and routed by an overarching broadcast master controller¹⁸. The psychological driver is not about "analog feel," but about packet loss prevention, sub-millisecond network latency, and seamless integration into dense IT infrastructures.

Absolute Mission Continuity and Disaster Recovery

For live television news and arena touring, silence is a catastrophic, career-ending failure. The Sonicview buyer is deeply motivated by disaster recovery topologies. The "dp" (dual power) models' redundant DC supply topology ensures that a physical power fault at the wall or within the primary internal PSU does not drop the broadcast⁶. Furthermore, TASCAM has segregated the 54-bit FPGA DSP engine from the graphical control surface operating system. This is a crucial psychological safety net for the buyer: in the highly unlikely event that the graphical user interface halts or requires a reboot, the underlying FPGA continues to pass audio, maintain

network clock, and process DSP uninterrupted⁷. This decoupling of control and computation is the hallmark of a system designed for uptime over all other considerations.

Algorithmic Workflow Automation

Broadcast buyers are motivated by the reduction of operator error and cognitive load. The native implementation of the Ember+ protocol is a massive technical draw. Ember+ allows newsroom control systems (NRCS), such as Viz Mosart or Nxtedition, to mathematically drive the console's faders and muting logic without human intervention, synchronizing audio perfectly with automated video rundowns²¹. Additionally, the Audio Follow Video (AFV) function, linked via GPIO tallies, allows the console to mathematically open faders based on the video switcher's logic, adhering to strictly defined rise times, hold times, and fall times²⁷. When human operators are involved in unscripted multi-panelist discussions, the DSP's Gain-Sharing Auto Mixer algorithm calculates real-time priority and weight values, dynamically suppressing unused microphones to prevent acoustic feedback and phase cancellation automatically²⁷. The motivation is to automate the mundane so the engineer can focus on the critical.

The 54-Bit FPGA Summing Advantage

Technical buyers inherently understand the mathematical limitations of older 32-bit or 40-bit fixed-point DSP chips, which can suffer from truncation errors and internal clipping when summing large channel counts. The motivation to adopt Sonicview lies in its 54-bit floating-point FPGA architecture. By dedicating 42 bits strictly to amplitude data and reserving 12 bits for headroom, the systems architect ensures that complex, heavy mix buses (summing 40+ high-transient signals simultaneously) will mathematically never clip the internal bus. This preserves absolute phase coherence and transient fidelity regardless of operator gain-staging errors upstream¹³.

STAGE 3 — Product Identity Narrative

Because the Sonicview architecture shares no DNA with the Model series' analog/digital hybridity, its identity narrative must aggressively discard any notion of analog regression or tactile nostalgia. Sonicview is not an analog board bolted to a digital recorder; it is a dedicated, ultra-high-speed computational matrix.

The core identity narrative for this production is: **"The Native IP DSP Core."**

This narrative strictly defines the console not as a traditional soundboard, but as a centralized processing node on a distributed digital network. The physical 100mm faders and capacitive VIEW touchscreens are conceptualized merely as the human-machine interface controlling a deterministic 54-bit FPGA engine that lives seamlessly on a Dante or SMPTE ST 2110 network¹⁴. In this narrative, the SB-16D stagebox is not just an accessory; it is an extension of the console's physical nervous system, connected via CAT5e STP cables¹⁰. The analog inputs (Class 1 HDIA) exist for one singular purpose: to capture acoustic waveforms with surgical, instrumentation-grade accuracy, digitize them via 32-bit ADCs at the very edge of the device, and surrender them immediately to the floating-point math of the FPGA¹. The identity is clinical, mathematical, and infinitely routable.

STAGE 4 — Feature Prioritization

To reward genuine engineering comprehension and align with the hardcore technical framing, features are prioritized based on their direct impact on system architecture, signal integrity, and broadcast determinism.

Narrative Rank	Technical Feature	Engineering Justification
1	54-Bit Floating-Point FPGA Engine	The mathematical heart of the console. Dictates the 2-sample (20.8μs) DSP latency and the infinite internal headroom of the mix bus, ensuring mathematically perfect summing ¹ .
2	Native Dante 64x64 & ST 2110 IP Topologies	Establishes the console as a broadcast AoIP network node. The ST 2022-7 packet redundancy and NMOS discovery elevate the unit from a live mixer to a broadcast routing hub ¹⁸ .
3	Class 1 HDIA Preamplifier Topology	The analog-to-digital gateway. The true instrumentation amplifier circuit design yields an exceptional -128 dBu EIN and +32 dBu headroom, capturing flawless transients prior to the 32-bit ADC ¹ .
4	Hardware & Software Disaster Recovery	The separation of the OS from the DSP engine (audio passes even if UI halts) combined with the "dp" models' dual-redundant AC/DC power supply ensures mission-critical uptime ⁶ .

5	Broadcast Automation Logic	Native implementation of Ember+ for newsroom automation control, alongside Audio Follow Video (AFV) tally logic and real-time Gain-Sharing Auto Mixers for unscripted environments ²² .
6	TASCAM VIEW HMI Architecture	The integration of multiple 7-inch touchscreens, custom motorized fader layers, User Defined Controls, and DCA spill capabilities to reduce operator cognitive load ¹ .

STAGE 5 — Visual Research and Cinematographic Grammar

The visual grammar must abandon the warm, tactile, macro-photography of analog potentiometers previously established for the Model series. Sonicview requires a clinical, high-tech, data-driven cinematographic approach that emphasizes the invisible movement of data.

- **Lighting and Palette:** Clinical, cold, and high-contrast. The lighting design should utilize sharp edge-lighting (cyan, deep blue, stark white) to highlight the metallic chassis, the matte finish of the 100mm fader tracks, and the glow of the capacitive screens. The environmental aesthetic should evoke a sterile server room or a broadcast Master Control Room.
- **The FPGA/Digital Focus:** Since the actual mixing happens invisibly in microchips, visual emphasis must be placed on the physical manifestations of digital control. Use high-frame-rate macro cinematography of the 100mm motorized faders snapping violently yet precisely to position during a Snapshot Recall, emphasizing mechanical speed and digital determinism.
- **Screen Legibility as Subject:** The multiple 7-inch VIEW touchscreens are primary subjects, not background elements. Cinematography must actively manage lens polarization and shutter angles to capture the RTA (Real-Time Analyzer) graphs, PEQ curves, and dynamic compression UIs with crisp, anti-aliased perfection. Screen reflections must be rigorously controlled.
- **I/O Topography:** Focus macro lenses on the locking mechanisms of the etherCON Dante ports, the DB25 GPIO pin arrays, and the BNC Word Clock terminals. This communicates broadcast-level integration. Extremely shallow depth of field should be used to isolate a

specific network port (e.g., Dante Primary) while the rest of the I/O bay falls into clinical, structured bokeh.

STAGE 6 — Motion Planning Concepts

To visualize the invisible digital architecture and mathematical superiority of Sonicview for the downstream motion design build, the following reusable visual concepts should be developed:

1. **The 54-Bit Summing Matrix:** A 3D motion concept visualizing 44 distinct audio waveforms (representing the internal input channels) entering a central computational node (the FPGA). The visualization should show the waveforms expanding vertically as they sum together. Crucially, they never clip against a digital ceiling, representing the 42-bit data + 12-bit headroom mathematics of the 54-bit float architecture.
2. **ST 2022-7 Redundant Packet Flow:** A network visualization showing the console and the SB-16D stagebox connected via dual Ethernet lines. A stream of digital data packets splits into two identical streams across Primary and Secondary Dante networks. Mid-stream, one path is artificially severed (representing a cable fault), but the destination node seamlessly receives the secondary packets with zero interruption, visualizing true network redundancy.
3. **Audio-Follow-Video (AFV) Tally Logic:** A split-screen graphic. On one side, a binary GPIO trigger flips from 0 to 1 (representing a camera tally activation). A graphical logic line connects this to the other side of the screen, where a digital fader mathematically calculates its rise time, hold time, and fall time, moving automatically on a parabolic curve based entirely on the hardware trigger.
4. **Class 1 HDIA Instrumentation Stage:** A schematic overlay positioned over physical footage of the XLR port. The camera pushes into a circuit diagram of the instrumentation amplifier, highlighting the extremely low noise floor (-128 dBu) and high Common-Mode Rejection Ratio (CMRR) actively filtering out electromagnetic interference before hitting the 32-bit ADC block.

STAGE 7 — Storytelling Arc

The narrative arc for this production is clinical, problem/solution-oriented, and rigorously technical, culminating in absolute engineering mastery and the validation of Shivansh Electronics as the integration partner.

- **Phase 1: The Broadcast Challenge (The Problem):** Establish the fragility and complexity of modern audio environments. Highlight the dangers of high channel counts, the mathematical threat of digital clipping on heavy summing buses, the disaster of packet loss in IP networks, and the catastrophic nature of power or OS failure in live broadcast.
- **Phase 2: The Computational Solution (The FPGA & HDIA):** Introduce the Sonicview core. Detail the acoustic capture via Class 1 HDIA instrumentation amps feeding 32-bit ADCs. Transition immediately into the processing core, explaining the 54-bit float FPGA engine. Establish the sheer mathematical superiority of the internal matrix and the imperceptible 2-sample latency.
- **Phase 3: Network Topologies & Automation (The Ecosystem):** Zoom out from the isolated console to the wider network fabric. Introduce Dante 64x64, ST 2110 broadcast

interoperability, and the SB-16D networked stageboxes. Detail how Ember+ automation and Gain-Sharing Auto Mixers allow the console to function autonomously in a broader broadcast ecosystem. Highlight the dual-power (dp) disaster recovery capability.

- **Phase 4: The HMI & Deployment (Control):** Bring the focus back to the human operator interacting with the machine. Demonstrate the VIEW touchscreens, custom motorized fader layers, DCA spill, and instantaneous Snapshot recalls.
- **Phase 5: Technical Validation (The Resolution):** Conclude with the strict validation of Shivansh Electronics as the authorized engineering partner capable of deploying, configuring, and supporting these rigorous, mission-critical network ecosystems.

STAGE 8 — Verified Technical Specification Master Table

Note: Downstream production systems must use these exact, verified figures. All DSP and network parameters apply equally to both XP and dp variants unless otherwise specified in the Power Topology section.

TASCAM Sonicview 16 & Sonicview 24 Specifications

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Core Specification	TASCAM Sonicview 16 (XP/dp)	TASCAM Sonicview 24 (XP/dp)	Verification Source
Mixing Engine Topology	54-bit floating-point FPGA	54-bit floating-point FPGA	VERIFIED ¹
Internal DSP Latency	2 samples (20.8 μs) @ 96kHz	2 samples (20.8 μs) @ 96kHz	VERIFIED ⁴
Analog-to-Analog Latency	0.51 ms	0.51 ms	VERIFIED ⁴
A/D - D/A Conversion	32-bit ADC / 24-bit DAC @ 96kHz	32-bit ADC / 24-bit DAC @ 96kHz	VERIFIED ²
Internal Input Channels	44 (40 Mono, 2 Stereo) + 4 FX Return	44 (40 Mono, 2 Stereo) + 4 FX Return	VERIFIED ²
Output Bus	22 Flex, Main L/R, 4	22 Flex, Main L/R, 4	VERIFIED ²

Architecture	FX Send (32 Total)	FX Send (32 Total)	
Analog Mic/Line Inputs	16 XLR (Ch 9-16 combo TRS)	24 XLR (Ch 17-24 combo TRS)	VERIFIED ¹
Analog Line Outputs	16 XLR	16 XLR	VERIFIED ¹
Preamp Architecture	Class 1 HDIA (Instrumentation Amp)	Class 1 HDIA (Instrumentation Amp)	VERIFIED ¹
Preamp EIN	-128 dBu or less	-128 dBu or less	VERIFIED ¹
Max Input Level (Mic/Line)	+32 dBu (Pad ON, Trim Min)	+32 dBu (Pad ON, Trim Min)	VERIFIED ²
Motorized Faders	16 Channel + 1 Master (100mm)	24 Channel + 1 Master (100mm)	VERIFIED ¹
VIEW Touchscreens	Two (7-inch Color Capacitive/Resistive)	Three (7-inch Color Capacitive/Resistive)	VERIFIED ¹
Native Network I/O	64x64 Dante (48kHz), 32x32 (96kHz)	64x64 Dante (48kHz), 32x32 (96kHz)	VERIFIED ²
USB Audio Interface	32-in/32-out (32-bit/96kHz)	32-in/32-out (32-bit/96kHz)	VERIFIED ²
SDXC Multitrack (IF-MTR32)	32-Track Direct-to-SD (included in XP)	32-Track Direct-to-SD (included in XP)	VERIFIED ²
Control Protocols	Ember+, NMOS IS-04/05, SNMP, GPIO	Ember+, NMOS IS-04/05, SNMP, GPIO	VERIFIED ²

Dimensions (W x H x D)	472.0 × 228.1 × 554.4 mm	690.8 × 228.1 × 554.4 mm	VERIFIED ²
Weight	13 kg	18 kg	VERIFIED ²
Power Supply (XP variants)	Single AC (100-240V)	Single AC (100-240V)	VERIFIED ²
Power Redundancy (dp variants)	AC + Redundant DC (PS-P2450)	AC + Redundant DC (PS-P2450)	VERIFIED ⁶

TASCAM SB-16D Stagebox Specifications

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Core Specification	TASCAM SB-16D	Verification Source
Form Factor	3U Rackmountable / Floor Stagebox	VERIFIED ⁸
Analog I/O	16 XLR Mic/Line In, 16 XLR Line Out	VERIFIED ⁸
Preamp Architecture	Class 1 HDIA (-128 dBu EIN, +32 dBu Max)	VERIFIED ⁸
A/D - D/A Conversion	32-bit / 96kHz	VERIFIED ⁸
Dante Network Integration	64x64 (48kHz) / 32x32 (96kHz)	VERIFIED ⁸
Network Connectors	Primary & Secondary etherCON	VERIFIED ⁸
Network Protocols Supported	Dante Domain Manager, AES67, ST 2110	VERIFIED ⁸
GPIO Interface	8-in/8-out (Expands to 16/16)	VERIFIED ⁸

	with console)	
Power Topology	Internal AC + Redundant DC Input	VERIFIED ⁸
Dimensions (W x H x D)	482.8 × 132 × 120 mm (Rackmount configuration)	VERIFIED ⁸
Weight	4.5 kg	VERIFIED ⁸

STAGE 9 — Voiceover Tone Research

The voiceover delivery must reject commercial enthusiasts entirely. The tone must be delivered with rigorous clinical precision, sounding like a senior systems architect addressing professional engineering peers. The delivery should be measured, data-heavy, and unyielding in its technical authority.

Direction 1: The Network Architect (Clinical, Data-Driven, Fast-Paced)

Sample: "Digital audio is no longer constrained by the physical limits of the chassis. The TASCAM Sonicview architecture abandons traditional DSP chips, centralizing all processing within a 54-bit floating-point FPGA engine. Operating natively at 96 kilohertz, this yields a continuous, imperceptible 20.8 microsecond internal latency. Outfitted with native ST 2022-7 Dante redundancy, AES67 interoperability, and native Ember+ logic for newsroom automation, the Sonicview operates not just as a mixing console, but as a fully deterministic node on your overarching IP fabric."

Direction 2: The Acoustic Physicist (Precision, Component-Focused, Deliberate)

Sample: "Absolute signal integrity begins at the analog boundary, long before the digital conversion. The Class 1 HDIA preamplifiers deploy a true instrumentation amplifier topology at the very first gain stage. Delivering an equivalent input noise of negative 128 dBu and an extreme +32 dBu of transient headroom, the acoustic waveform is mathematically preserved, rejecting electromagnetic interference via massive common-mode rejection, before it ever reaches the 32-bit analog-to-digital converter."

Direction 3: The Master Control Engineer (Mission-Critical, Authoritative, Stern)

Sample: "In live broadcast and high-stakes touring, catastrophic failure is not a metric we tolerate. The Sonicview 'dp' series physically isolates the 96 kilohertz audio DSP engine from the graphical operating system, ensuring continuous, uninterrupted signal flow even if the user interface were to inexplicably halt. Paired with dual-redundant AC and DC power supplies utilizing the PS-P2450 DC adapter, absolute disaster recovery is physically engineered into the console's fundamental DNA."

STAGE 10 — Typography, Information Hierarchy, and Mandatory Branding/Contact Requirements

Because commercial pricing language, MOP figures, and currency amounts are strictly

excluded by the project constraints, numerical authority within the visual frame must be established through elite engineering statistics. These verified figures will serve as "Hero Typography," anchoring the visual composition where price tags or commercial copy would traditionally sit.

Hero Typography Anchors

- **54-BIT / 96 kHz** (Subtext: Floating-Point FPGA Engine)
- **-128 dBu EIN** (Subtext: Class 1 HDIA Instrumentation Topology)
- **20.8 μ s** (Subtext: 2-Sample DSP Latency)
- **ST 2110 / ST 2022-7** (Subtext: Broadcast IP Redundancy)
- **64x64 AoIP** (Subtext: Deterministic Dante Integration)

Information Hierarchy

When technical figures are presented on screen, the downstream visual system must adhere to the following strict hierarchy to ensure engineering legibility:

1. **Macro Hardware Shot:** Deep focus on specific hardware correlating to the data (e.g., etherCON locking ports for Dante specs, or motorized faders for HMI specs).
2. **Hero Typography Overlay:** Large, bold, wide-tracked numerical figures (e.g., **54-BIT**).
3. **Subtext Definition:** Thin, technical monospaced font defining the metric immediately below the hero text (e.g., INTERNAL SUMMING ARCHITECTURE).
4. **Data Block:** A small, transparent digital bounding box containing 3-4 bulleted sub-specs to reward pausing the video (e.g., 32-Bit ADC | +32 dBu Headroom | 42-Bit Data / 12-Bit Margin).

Mandatory Branding & Contact Integration

The following elements are mandatory. They must be omnipresent in the lower-thirds or featured in dedicated end-cards. They must be set in a clean, high-contrast, technical sans-serif font, strictly adhering to the "Authorized Partner" terminology, and never obscuring the core technical data overlays.

- **Logo:** [Shivansh Electronics Logo] – Persistent on-screen watermark, recommended in the lower right quadrant.
- **Designation:** Authorized Partner of TASCAM (*Strictly this phrasing; absolutely no use of 'Authorized Distributor' or territory-specific language*).
- **Digital Footprint:**
 - www.shivanshelectronics.in
 - [instagram.com/@shivanshelectronics.in](https://www.instagram.com/@shivanshelectronics.in)
 - [facebook.com/@shivanshelectronics.in](https://www.facebook.com/@shivanshelectronics.in)
 - [youtube.com/@shivanshelectronics-in](https://www.youtube.com/@shivanshelectronics-in)
- **Direct Contact (Voice Call & WhatsApp Supported):**
 - +91 98316 62458
 - +91 91477 00677
 - +91 89818 07755

Confirmation: Scope ends explicitly at Stage 10. No Stage 11 or video-generation prompts of any kind have been included in this deliverable, reserving full creative autonomy for the

downstream visual-generation pipeline in accordance with established constraints.

Works cited

1. TASCAM Sonicview 24XP / TASCAM Sonicview 24dp - United States, https://tascam.com/us/product/sonicview_24xp
2. TASCAM Sonicview 24 / TASCAM Sonicview 24dp | Digital Mixer, https://tascam.jp/int/product/sonicview_24
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